

## Prevalence and molecular characterization of *Mycobacterium tuberculosis* resistance to aminoglycosides in the Northeast of Iran

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### Summary

**Background:** Tuberculosis is an important cause of death with an increased antibiotic resistance. Kanamycin and amikacin are among the most effective aminoglycosides for multidrug-resistant strains. We investigated the resistance to these two antibiotics in 111 susceptible and 9 resistant specimens.

**Materials and Methods:** Phenotypic method was used to assess the resistance, PCR method was performed for *rrs* and *eis* genes and the results confirmed by sequencing.

**Results:** No resistance against kanamycin and amikacin was observed in susceptible specimens and only three resistant samples had such resistance. Two had mutations at A1401G nucleotide, and the nucleotide change of G→A was observed at 1484 position in one sample. Mutations at

A514C, A907C, G1332A positions were observed in one sample. *eis* gene sequencing revealed that one sample had mutation in promoter region at A-13G position.

Conclusion: The most important mutation was at nucleotide A1401G in *rrs* gene and a low prevalence of aminoglycoside resistance was observed in northeastern Iran.



### Key words

*Tuberculosis, Resistance, Aminoglycosides, MDR-TB*

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## 1. Introduction

Tuberculosis (TB) is an infectious disease caused by *Mycobacterium tuberculosis* and is among the ten major causes of death due to infectious agents.<sup>1-4</sup> According to the World Health Organization report in 2018, 10 million people were infected, and 558,000 of them were resistant to rifampicin (RR-TB), and of these, 82% had multidrug-resistant tuberculosis (MDR-TB).<sup>5</sup> The emergence of drug resistance among *Mycobacterium* strains has caused a global concern. Multidrug-resistant tuberculosis defined as resistance to two drugs, isoniazid and rifampicin, and extensively drug-resistant tuberculosis (XDR-TB) defined as resistance to at least one of the injectable drugs such as amikacin and kanamycin and any of fluoroquinolones. Despite the appropriate tuberculosis control program in Iran, drug resistance has been observed among infected people. Rapid and timely diagnosis is an essential factor in the prevention of resistant tuberculosis. Molecular detection needs less time and has particular importance in the detection of drug resistance. In the suggested regimen by WHO, this group is the second line of tuberculosis treatment including injectable drugs like aminoglycosides and polypeptides.<sup>6</sup> Amikacin and kanamycin are two aminoglycosides

and have shown good results in the treatment of tuberculosis. Aminoglycosides irreversibly bind to the 16s rRNA in the 30s ribosomal region, and subsequently, the protein synthesis is disrupted and inhibited. Resistance to these drugs is mainly due to the mutation in the *rrs* gene encoding 16s rRNA, and promoter region of *eis* (enhanced intracellular survival) gene.<sup>7</sup> Therefore, due to the importance of drug resistance and the gradual spread of this resistance among second-line agents, this study was conducted to assess the prevalence of resistance to amikacin and kanamycin as the injectable drugs and their mutations involved in the onset of resistance.

## 2. Materials and methods

### 2.1 Data collection and sample preparation

In this study, 111 susceptible and 9 MDR clinical isolates of *Mycobacterium tuberculosis* were collected. Samples were obtained from patient's sputum who had been referred to tuberculosis reference laboratories of northeast of Iran during January to August 2017. The study was approved by the Institutional ethical committee (No. 950690 in Mashhad University of medical sciences). Population characteristics is pre-

sented in the Table 1. Characteristics of patients with MDR-TB and their pattern of drug resistance is provided in the Table 2.

Sputum samples were decontaminated and digested by Petroff's method.<sup>8</sup> Then, the smear was prepared from each sample. Prepared smears were stained by the Ziehl-Neelsen method. Finally, the stained smear was examined by microscope to detect acid-fast bacilli. *Mycobacterium tuberculosis* H37Rv was used as the quality control for phenotypic and genotypic tests.

## 2.2 Drug Susceptibility Test

After growth and confirmation of *Mycobacterium tuberculosis* colonies on Lowenstein-Jensen (LJ) medium, a drug susceptibility test was performed. Resistance to each of amikacin and kanamycin antibiotics were examined on LJ medium containing each antibiotic, by proportional method.<sup>9</sup> The critical concentration of 30 µg/ml used for kanamycin and amikacin, according to the WHO suggestion.<sup>10</sup>

## 2.3 DNA extraction and PCR amplification

In order to purify the genomic DNA of resistant samples, the boiling method was used. Colonies were suspended in 400 µl of 1X TE buffer (10 mM Tris-HCl [PH=7], 1 mM EDTA) and boiled for 30 minutes. Then the suspension was centrifuged at 12000 rpm for 20 minutes, and the supernatant which contained nucleic acids was transferred to a new microtube and stored at -20°C.

To detect two regions of *rrs* gene (500 and 1400), 1119 bp fragment was amplified. Primers were designed and examined by gene runner (version 6.5) and beacon designer (version 8.10) softwares. Also, a 602 bp fragment of *eis* gene was amplified by primers selected from the study by Joao Perdiago et al.<sup>11</sup> The primers and amplicon size are provided in Table 3.

Extracted DNA was used as the template for PCR reaction for *rrs* and *eis* genes. PCR for *rrs* gene was performed at a final volume of 25 µl containing 2.5 µl PCR buffer, 1.5 mM MgCl<sub>2</sub>, 0.5 µl DMSO, 0.3 µl *Taq* DNA polymerase, 200 µM dNTP, 0.5 µM of each forward and

**Table 1** Population characteristics of 120 studied cases

|       | New cases (untreated) | Recurrent TB | Treated TB | Treatment failure TB | MDR) |
|-------|-----------------------|--------------|------------|----------------------|------|
| Men   | 51                    | 5            | 5          | 2                    | 7    |
| Women | 33                    | 11           | 3          | 1                    | 2    |

**Table 2** Characteristics of patients with MDR-TB and their patterns of drug resistance

| Patients No. | Drug-resistant pattern |     |     |     |     | Sex | Age | Nationality | Previous TB treatment) |
|--------------|------------------------|-----|-----|-----|-----|-----|-----|-------------|------------------------|
|              | INH                    | RIF | EMB | KAN | AMK |     |     |             |                        |
| MTB 1        | R                      | R   | R   | R   | R   | M   | NA  | Iranian     | NA                     |
| MTB 2        | R                      | R   | R   | R   | R   | M   | 35  | Iranian     | YES                    |
| MTB 3        | R                      | R   | R   | R   | R   | M   | NA  | Migrant     | NA                     |
| MTB 4        | R                      | R   | R   | S   | S   | F   | 55  | Iranian     | YES                    |
| MTB 5        | R                      | R   | R   | S   | S   | M   | 60  | Iranian     | YES                    |
| MTB 6        | R                      | R   | S   | S   | S   | F   | 73  | Migrant     | YES                    |
| MTB 7        | R                      | R   | R   | S   | S   | M   | 23  | Iranian     | YES                    |
| MTB 8        | R                      | R   | R   | S   | S   | M   | NA  | Migrant     | NA                     |
| MTB 9        | R                      | R   | R   | S   | S   | M   | NA  | Iranian     | YES                    |

*R: resistant, S: sensitive, M: male, F: female, NA: not available, INH: isoniazid, RIF: rifampicin, EMB: ethambutol, KAN: kanamycin, AMK: amikacin*

**Table 3** Primers used for amplification of *rrs* and *eis* gene

| Gene       | Primer  | Sequences            | Amplicon size (bp) |
|------------|---------|----------------------|--------------------|
| <i>rrs</i> | Forward | GGATGACGGCCTTCGGGTTG | 1119               |
|            | Reverse | GATCCAGCCGCACCTTCCG  |                    |
| <i>eis</i> | Forward | GCCATGGGACCGGTACTTGC | 602                |
|            | Reverse | GTAGATGCCGCCCTCGCTAG |                    |

**Table 4** Analysis of *rrs* and *eis* genes and their mutations in 3 resistant samples based on sequencing results, R: Resistant

| MTB clinical isolates | Aminoglycoside Resistance |           | <i>rrs</i> mutation (nucleotide change) | <i>eis</i> mutation |
|-----------------------|---------------------------|-----------|-----------------------------------------|---------------------|
|                       | Amikacin                  | Kanamycin |                                         |                     |
| MTB 1                 | R                         | R         | A514C<br>G1332A<br>G1484A               | A-13G               |
| MTB 2                 | R                         | R         | A1401G                                  | No mutation         |
| MTB 3                 | R                         | R         | A907C<br>A1401G                         | No mutation         |

reverse primers and 1 µl DNA. The cycling program was as the initial denaturation at 95°C for 5 minutes followed by 35 cycles of denaturation at 95°C for 1 minute, annealing at 64°C for 1 minute, extension at 72°C for 1 minute, and a final extension at 72°C for 10 minutes.

PCR reaction for *eis* gene was prepared at the final volume of 20 µl containing 2 µl PCR buffer, 2 mM MgCl<sub>2</sub>, 200 µM dNTP, 0.2 µl *Taq* DNA polymerase, 0.5 µl DMSO, 0.5 µM of each forward and reverse primers and 2 µl DNA. The cycling program included initial denaturation at 95°C for 5 minutes followed by 35 cycles of denaturation at 95°C for 35 seconds, annealing at 62°C for 35 seconds, extension at 72°C for 1 minute and a final extension at 72°C for 10 minutes. The PCR products were analyzed by gel electrophoresis.

#### 2.4 Sequencing

Sequencing was performed on purified PCR products by Korean Bioneer Company to detect mutations associated with resistance. The results were compared to the wild-type H37Rv, and mutations related to resistance were analyzed by chromas (version 2.6.5), BioEdit (version 7.2.6) and Mega (version 5.10) softwares.

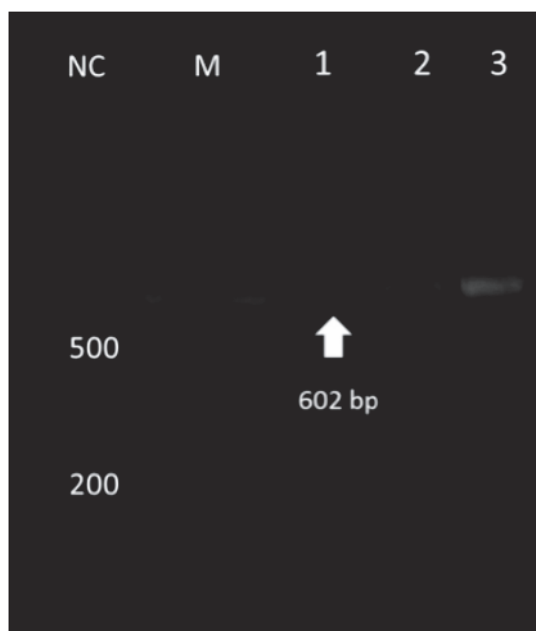
### 3. Results

#### 3.1 Phenotypic analysis results (Drug Susceptibility Test)

Of the 111 susceptible *Mycobacterium tuberculosis* tested clinical samples, no drug resistance to kanamycin or amikacin were observed, and only 3 out of 9 MDR samples were resistant to kanamycin and amikacin.

#### 3.2 Analysis of *rrs* and *eis* genes associated with resistance

3 out of 9 MDR samples were resistant to kanamycin and amikacin. Sequencing results of *rrs* gene were analyzed, and mutations related to aminoglycoside resistance were determined. All three samples had mutations in *rrs* gene at different positions. Nucleotide change at position A1401G was observed in two samples and mutations at A907C, and A514C position was also observed in two samples. Furthermore, nucleotide changes were detected at 1484 and 1332 positions. Analysis of *eis* gene revealed that only one of the samples had a mutation in the promoter region at A-13G position. The results of analysis for the three resistant samples and their mutations are provided in



**Figure 1**

*eis* gene PCR results for resistant clinical isolates showing 602 bp band. Lane 1 to 3: resistant clinical isolates. NC: negative control, M: 100 bp DNA size marker



**Figure 2**

*rrs* gene PCR results for resistant clinical isolates showing 1100 bp band. Lane 1 to 3: resistant clinical isolates. NC: negative control. M: 100 bp DNA size marker

Table 4. The results of PCR test for *eis* and *rrs* genes are provided in the Figures 1 and 2, respectively.

#### 4. Discussion

Emergence of MDR and XDR tuberculosis caused a global concern in the control of the disease. Therapeutic regimen used in the treatment of tuberculosis is the main factor in the occurrence of this resistance, although rapid and accurate diagnosis also has an important role.

Phenotypic drug susceptibility tests are important in the diagnosis of drug resistance. Aminoglycosides are among the most effective drugs in the tuberculosis treatment.<sup>7</sup> Resistance to these drugs is increasing gradually. Identification of mutations involved in resistance has an important role in the diagnosis of resistant strains. In this study, we investigated two *rrs* and *eis* genes and their mutations related to aminoglycoside resistance.

The prevalence of resistance to injectable drugs was higher among MDR strains, as none of the susceptible *M. tuberculosis* samples were resistant to kana-

mycin and amikacin. These results indicated the low prevalence of resistance among susceptible strains of *M. tuberculosis* in Iran.

Changes of nucleotides in different positions of the *rrs* gene cause resistance, and the most mutations occur in 500, 900, and 1400 regions. The most common mutation is at A1401G and less frequent at C1402T and G1484T nucleotides. However, few aminoglycoside-resistant strains have shown mutations at 514, 517, and 907 nucleotides.<sup>12</sup> In addition to this gene, the mutation in the *eis* promoter region also causes low levels of resistance to kanamycin and amikacin. The nucleotide change in positions G-10A, C-12T, C-14T and G-37T of this gene, results in aminoglycoside resistance.

The increased frequency of mutation was located at A1401G nucleotide which was related to a high level of resistance to amikacin and kanamycin.<sup>13</sup> In this study, similar to other reports, mutation in the listed nucleotides has been detected. Although nucleotide change C→T at 1402 position has been observed in amikacin and kanamycin resistant strains in other studies, none of the resistant samples showed mutation at this nucleotide in our study.

Mutation at 1484 nucleotide is also related to kanamycin and amikacin resistance, but its prevalence is lower than other mutations.<sup>14</sup> Regarding this, only one of the samples showed this mutation in this study which indicated a low prevalence of this mutation among the strains. In addition to the mentioned positions, a mutation at 514 and 907 nucleotides was also associated with kanamycin and amikacin resistance with a lower frequency, and this mutation has been detected only in some limited reports.<sup>11,12</sup> Our results revealed that only one sample had nucleotide change at the A514C position and also another sample had a mutation at A907C nucleotide. However, despite the low prevalence of these two mutations among resistant strains, further investigation is needed to determine the definite effect of these mutations in the resistance to the aminoglycosides.

Another nucleotide change which is related to aminoglycoside resistance is located at 1332 position. This nucleotide change occurs in rare cases, and reports on this mutation are not as much as others. Also, only one sample showed this mutation in this study.

Mutations at *eis* gene are also associated with aminoglycoside resistance. Generally, the mutation at promoter regions of this gene causes more resistance to kanamycin and less to amikacin. The prevalence of this mutation in contrast to *rrs* gene is lower throughout other countries. Mutations which occurred in the promoter region of *eis* gene are mostly located at G-10A, C-12T, C-14T and G-37T positions.<sup>6,15</sup> Due to our sequencing results, only one sample had a mutation in the promoter region at A-13G position which was different from other reports. This nucleotide change may suggest a new pattern of drug resistance in *eis* gene among *M. tuberculosis* strains in Iran. However, more studies are needed to ensure the role of this nucleotide change in aminoglycoside resistance.

In this study, the most frequent mutation related to aminoglycoside resistance was located at 1401 position, similar to other studies. This issue indicated that the resistance pattern among *M. tuberculosis* strains in Iran was the same as other countries. Despite the low frequency of mutations at A514C and A907C positions, our results suggested that further investiga-

tion is needed to confirm the role of these mutations in the resistance to amikacin and kanamycin. Moreover, the mutation at the promoter region of *eis* gene is also related to aminoglycoside resistance. Therefore, the evaluation of phenotypic susceptibility tests and performing molecular methods is very important in the detection of drug resistance among MDR strains. However, as we had some limitations such as small sample size and methods restrictions, further studies should be performed on a larger sample size and MIC is suggested to be determined.

## 5. Conclusion

Emergence of drug resistance among *Mycobacterium* strains with its increasing rate has made a concern all over the world. Rapid and accurate detection methods can help to control and prevent this resistance. According to our results, mutations related to aminoglycoside resistance mainly occur at different positions of *rrs* gene, however this mutation mostly places at the 1401 nucleotide. Furthermore, a mutation at *eis* promoter region is also related to resistance with low frequency. It can be concluded that aminoglycoside resistance was low in our study region and also the pattern of resistance in Iran was the same as other countries.

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### Authors' contributions

FA, EA, KG conceived and designed the study. All authors contributed to the study protocol and analyzed the data. FA, KG drafted the manuscript, critically reviewed it and prepared the final version. All authors read and approved the final manuscript

### Conflict of Interest

The authors declare that they have no conflict of interest.





## Περίληψη

### Επιπολασμός και μοριακός χαρακτηρισμός της αντοχής του *Mycobacterium tuberculosis* στις αμινογλυκοσίδες στο ΒΑ Ιράν

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**Σκοπός:** Η φυματίωση είναι μια σημαντική αιτία θανάτου, ενώ παράλληλα ο υπεύθυνος μικροοργανισμός, *M. tuberculosis*, παρουσιάζει αυξημένη αντοχή στα χορηγούμενα αντιφυματικά. Η καναμυκίνη και η αμικασίνη είναι από τις πλέον αποτελεσματικές αμινογλυκοσίδες για τα πολυανθεκτικά *M. tuberculosis* (MDR-TB). Μελετήθηκε η αντοχή στα δύο αυτά αντιβιοτικά σε 111 ευαίσθητα και 9 ανθεκτικά στελέχη *M. tuberculosis*.

**Υλικό-Μέθοδος:** Η αντοχή σε καναμυκίνη και η αμικασίνη ελέγχθηκε με την χρήση φαινοτυπικής μεθόδου, ενώ για την μελέτη των υπεύθυνων για την αντοχή στα ανωτέρω αντιβιοτικά γονιδίων *rrs* και *eis* χρησιμοποιήθηκε η μέθοδος της PCR. Οι μεταλλαγές των γονιδίων επιβεβαιώθηκαν με αλληλούχιση.

**Αποτελέσματα:** Τα δείγματα των ευαίσθητων στελεχών *M. tuberculosis* δεν παρουσίασαν αντοχή σε καναμυκίνη και αμικασίνη, ενώ μόνο τρία από τα MDR-TB είχαν αντοχή στις υπό έλεγχο αμινογλυκοσίδες. Σε δύο εξ αυτών εντοπίστηκαν μεταλλαγές στο νουκλεοτίδιο A1401G, ενώ σε ένα δείγμα παρατηρήθηκε αλλαγή νουκλεοτιδίου του G → A στη θέση 1484. Μεταλλαγές στις θέσεις A514C, A907C, G1332A παρατηρήθηκαν σε ένα δείγμα. Η αλληλούχιση του γονιδίου *eis* αποκάλυψε σε ένα δείγμα μεταλλαγή στην περιοχή του υποκινητή στη θέση A-13G.

**Συμπέρασμα:** Παρατηρήθηκε χαμηλός επιπολασμός αντοχής στις αμινογλυκοσίδες σε στελέχη *M. tuberculosis* από το ΒΑ Ιράν, ενώ η σημαντικότερη μεταλλαγή που παρατηρήθηκε ήταν στο νουκλεοτίδιο A1401G του γονιδίου *rrs*.



#### Λέξεις κλειδιά

Φυματίωση, μικροβιακή αντοχή, αμινογλυκοσίδες, πολυανθεκτική φυματίωση (MDR-TB)



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